

Peichun Li

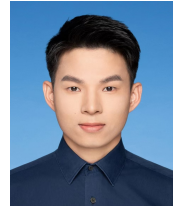
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I am a third-year Ph.D. student in Computer Science at the **University of Macau**, Macao SAR, China. Prior to this, I earned both my B.Eng. in Electronic Information Science and Technology and my M.Sc. in Control Science and Engineering at **Guangdong University of Technology**, Guangzhou, China.

My research focuses on **edge computing** and **distributed learning**, with a specific emphasis on **federated learning**, **collaborative large model deployment**, and **multi-agent AI systems** across edge networks.

Education

- 2024 – Present ■ **Ph.D. Student, University of Macau**
Major in *Computer Science*.
Supervisor: Prof. Yuan Wu.
- 2018 – 2021 ■ **M.Sc., Guangdong University of Technology**
Major in *Control Science and Engineering*.
Supervisor: Prof. Rong Yu.
The recipient of *National Scholarship for Graduate Students*.
- 2014 – 2018 ■ **B.Eng., Guangdong University of Technology**
Major in *Electronic Information Science and Technology*.
Supervisor: Prof. Rong Yu.
The recipient of *National Scholarship for Undergraduates*.

Selected Publications

- 1 **Peichun Li**, L. Qian, H. Zhang, Z. Shi, and Y. Wu, “Enabling multi-agentic AI in future wireless networks: An end-edge-cloud approach,” *IEEE Network*, 2026, early access (**JCR Q1**).
- 2 **Peichun Li**, H. Zhang, Y. Wu, L. Qian, R. Yu, D. Niyato, and X. Shen, “Filling the missing: Exploring generative AI for enhanced federated learning over heterogeneous mobile edge devices,” *IEEE Transactions on Mobile Computing*, vol. 23, no. 10, pp. 10 001–10 015, 2024 (**CCF A & JCR Q1**).
- 3 **Peichun Li**, H. Dong, L. Qian, S. Zhou, and Y. Wu, “FlexGen: Efficient on-demand generative AI service with flexible diffusion model in mobile edge networks,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 2, pp. 961–973, 2025 (**JCR Q1**).
- 4 **Peichun Li**, G. Cheng, X. Huang, J. Kang, R. Yu, Y. Wu, and M. Pan, “AnycostFL: Efficient on-demand federated learning over heterogeneous edge devices,” in *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications*, New York Area, USA, 2023 (**CCF A**).
- 5 **Peichun Li**, G. Cheng, X. Huang, J. Kang, R. Yu, Y. Wu, M. Pan, et al., “Snowball: Energy efficient and accurate federated learning with coarse-to-fine compression over heterogeneous wireless edge devices,” *IEEE Transactions on Wireless Communications*, vol. 22, no. 10, pp. 6778–6792, 2023 (**JCR Q1**).
- 6 **Peichun Li**, Y. Zhong, C. Zhang, Y. Wu, and R. Yu, “FedRelay: Federated relay learning for 6g mobile edge intelligence,” *IEEE Transactions on Vehicular Technology*, vol. 72, no. 4, pp. 5125–5138, 2023 (**JCR Q1**).
- 7 **Peichun Li**, L. Qian, D. Niyato, S. Mao, and Y. Wu, “Toward resource-efficient collaboration of large ai models in mobile edge networks,” *IEEE Network*, vol. 40, no. 3, pp. 51–59, 2026 (**JCR Q1**).
- 8 **Peichun Li**, G. Cheng, J. Kang, R. Yu, L. Qian, Y. Wu, and D. Niyato, “FAST: Fidelity-adjustable semantic transmission over heterogeneous wireless networks,” in *ICC 2023 - IEEE International Conference on Communications*, 2023, pp. 4689–4694.

- 9 **Peichun Li**, X. Huang, M. Pan, and R. Yu, “FedGreen: Federated learning with fine-grained gradient compression for green mobile edge computing,” in *2021 IEEE Global Communications Conference (GLOBECOM)*, 2021, pp. 1–6.
- 10 W. Guan, **Peichun Li**, H. Zhang, and Y. Wu, “Integrating generative AI with network digital twin for 6g: An edge-cloud collaborative approach,” *IEEE Communications Magazine*, vol. 63, no. 11, pp. 181–187, 2025 (**JCR Q1**).
- 11 T. Wang, **Peichun Li**, P. Feng, X. Wei, L. Qian, and Y. Wu, “Compression meets security: Low-complexity linear collaborative federated learning with enhanced accuracy,” *IEEE Transactions on Mobile Computing*, vol. 24, no. 10, pp. 10 606–10 621, 2025 (**CCF A & JCR Q1**).
- 12 R. Yu (Master’s Supervisor) and **Peichun Li**, “Toward resource-efficient federated learning in mobile edge computing,” *IEEE Network*, vol. 35, no. 1, pp. 148–155, 2021 (**JCR Q1**).
- 13 N. Huang, **Peichun Li**, L. Qian, Y. Ren, and Y. Wu, “Integrated sensing and communication for satellite–terrestrial integrated network with multi-access mobile edge computing,” *IEEE Internet of Things Journal*, vol. 13, no. 3, pp. 4099–4111, 2026 (**JCR Q1**).

Awards & Patents

- 2025  **Best Paper Award**, The 4th International Conference on Artificial Intelligence, Internet of Things and Cloud Computing Technology (AloTC). For the paper “Low-Altitude Task Offloading for UAV-USV Collaboration in Marine Edge-Enabled Networks.”
- 2025  **Exemplary Reviewer**, IEEE Transactions on Network Science and Engineering (TNSE). Recognized by IEEE Communications Society for contributions to peer review.
- 2024  **Excellent Paper Award (First Prize)**, Guangdong–Hong Kong–Macao Greater Bay Area Society for Artificial Intelligence and Automation. For the paper “Snowball: Energy Efficient and Accurate Federated Learning with Coarse-to-Fine Compression over Heterogeneous Wireless Edge Devices,” published in IEEE Transactions on Wireless Communications.
- 2022  **Granted Chinese Invention Patent**, “Resource Scheduling Method, Equipment, and Storage Medium.” Patent No.: CN110413396A. **First inventor**; granted on February 15, 2022.
- 2021  **Granted Chinese Invention Patent**, “Method and Equipment for Image Fusion in Vehicular Networks.” Patent No.: CN110363735B. **Second inventor** (master’s supervisor as first inventor); granted on August 13, 2021.

Technical Skills and Project Experience

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| Systems |  Hands-on experience with Linux and NVIDIA computing platforms, including Jetson Nano, Xavier, Orin, and DGX Spark. |
| Programming |  Strong programming skills in Python, C++, MATLAB, and Java; proficient in $\LaTeX$ for academic writing. |
| Research Projects |  Experience in preparing research grant proposals. Contributed to the design and implementation of a heterogeneous edge computing platform. |
| Software |  Developed a WeChat Mini Program serving graduate students across the university, with over 10,000 users. |

Academic Services

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| TPC Member |  IEEE GLOBECOM, HPCC, IEEE ICC, IEEE INFOCOM ICCN, and IEEE VTC. |
| Reviewer |  IEEE TMC, IEEE TWC, IEEE Network, IEEE TVT, IEEE TNSE, IEEE TCCN, IEEE IoTJ, et al. |